University of Wah

Department of Computer Science

Submit to:

Ms. Hassan Shafaqat

Submit by:

Muhammad Talha(101)
Muhammad Rafay Ali(063)
Jawad Hussain(108)

Course:

Computer Network_Lab - CS-251L

Reg No:

UW-24-CS-BS-101
UW-24-CS-BS-108
UW-24-CS-BS-063

Section:

BS-CS-3rd-B

1. Project Title

Software House Network

2. Problem Statement

Modern software houses consist of multiple departments such as Testing, Development, HR, Finance, and Management. When all systems operate on a flat network, it leads to major problems including poor performance due to broadcast traffic, lack of security between departments, limited scalability, and inefficient network management.

This project addresses the challenge of **network segmentation, security, and efficient communication** within a software house environment. Without proper logical separation, sensitive data (e.g., finance or HR information) can be exposed, and troubleshooting becomes complex as the organization grows.

3. Proposed Solution

The proposed solution is to design a **VLAN-based LAN architecture** where each department is assigned a separate VLAN. Inter-VLAN communication is enabled through a **Layer 3 routing device (Router-on-a-Stick)**. Centralized switches connect all departments, while servers and management systems are placed in dedicated VLANs for controlled access.

This approach improves performance, security, scalability, and administrative control while maintaining cost efficiency.

4. Objectives

- To design a structured VLAN-based network for departmental separation.
- To implement inter-VLAN routing for controlled communication between departments.
- To improve network security by isolating sensitive departments.
- To reduce broadcast traffic and improve overall network performance.
- To simulate and validate the network using Cisco Packet Tracer.

5. Network Environment and Tools

- **Network Type:** Local Area Network (LAN)
- **Architecture:** VLAN-based switched network with centralized routing
- **Tools/Software:**
    - Cisco Packet Tracer
    - Cisco 2960 Switches
    - Cisco Router (Inter-VLAN Routing)
- **Protocols & Technologies:**
    - VLAN (IEEE 802.1Q)

- Trunking
- IP Addressing & Subnetting

6. Methodology

1. Analyze software house departmental requirements.
2. Design logical VLAN structure for each department.
3. Assign IP addressing schemes to each VLAN.
4. Configure VLANs, access ports, and trunk ports on switches.
5. Implement inter-VLAN routing using a router.
6. Configure end devices and servers.
7. Test connectivity, isolation, and performance using simulation.
8. Verify results through ping and traffic testing.

7. Expected Outcomes

- Reduced broadcast traffic and improved network efficiency.
- Secure isolation of departmental data.
- Successful inter-department communication where required.
- Scalable network design suitable for organizational growth.
- Improved manageability and troubleshooting capabilities.

8. Applications

- Software houses and IT companies
- Corporate offices with multiple departments
- Educational institutions
- Small to medium enterprise (SME) networks
- Data-sensitive organizations requiring network segmentation

9. Conclusion

This project demonstrates a practical and scalable networking solution for a real-world software house environment. By implementing VLANs and inter-VLAN routing, the network becomes more secure, efficient, and manageable. The proposed design is cost-effective, easy to expand, and aligns with industry-standard networking practices, making it highly suitable for enterprise-level deployment.